

EE2703: Assignment 8

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1 Introduction

In this assignment, we shall look at DFTs, using the `numpy.fft` toolbox.

2 The Accuracy of the `numpy.fft` Package

To check how powerful and accurate the package is, we shall take the DFT of a sequence of random numbers, then take its IDFT and compare the two, as to how close they are. This is done using the commands `np.fft.fft` and `np.fft.ifft`. We also have to use `np.fft.fftshift` to shift the $[\pi, 2\pi]$ portion of the DFT to the left as it represents negative frequency, i.e., $[-\pi, 0]$.

The following piece of code accomplishes this task:

The error came out to be of the order of 10^{-15} . When the two sequences `xOriginal` and `xComputed` are plotted together, this is the result:

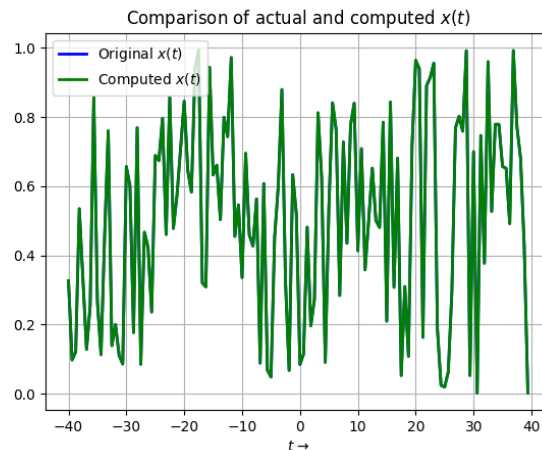
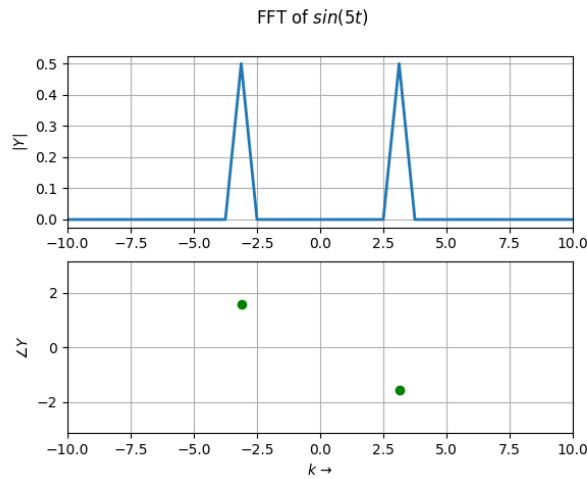


Figure 1: Comparison of the true and computed values of the random sequence

The two sequences overlap and cannot be differentiated !! This shows that the `numpy.fft` package is very accurate.

3 Spectrum of $\sin(5t)$

We calculate the DFT of $f(t)$ by the method mentioned in the above section. Then, we plot the phase and magnitude of the DFT and the following is obtained for $\sin(5t)$:

Figure 2: Spectrum of $\sin(5t)$

This is expected, because:

$$\sin(5t) = \frac{1}{2j}(e^{5jt} - e^{-5jt}) \quad (1)$$

So, the frequencies present in the DFT of $\sin(5t)$ are $\omega = \pm 5 \text{ rad/sec}$, and the phase associated with them is $\phi = \pm \frac{\pi}{2} \text{ rad/sec}$ respectively. This is exactly what is shown in the above plot.

4 Amplitude Modulation with $(1 + 0.1\cos(t))\cos(10t)$

We have,

$$(1 + 0.1\cos(t))\cos(10t) = \frac{1}{2}(e^{10jt} + e^{-10jt}) + 0.1 \cdot \frac{1}{2} \cdot \frac{1}{2}(e^{11jt} + e^{-11jt} + e^{9jt} + e^{-9jt}) \quad (2)$$

Writing $(1 + 0.1\cos(t))\cos(10t)$ in a different form as shown in (??), we observe that the frequencies present in the signal are $\omega = \pm 10 \text{ rad/sec}$, $\omega = \pm 11 \text{ rad/sec}$ and $\omega = \pm 9 \text{ rad/sec}$. Thus we expect the DFT also to have non-zero magnitudes only at these frequencies.

Plotting the DFT using the `numpy.fft` package, we get:

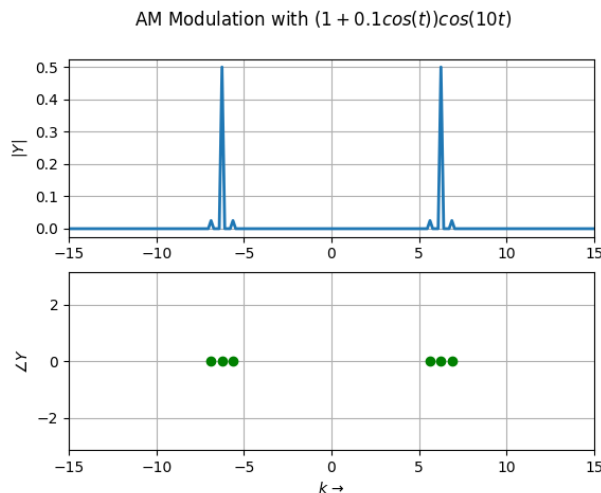
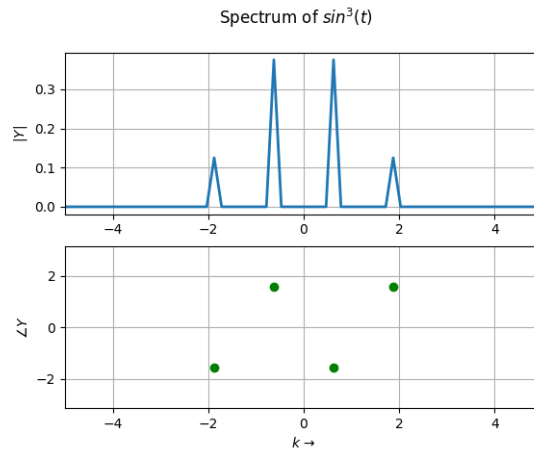
Figure 3: DFT of $(1 + 0.1\cos(t))\cos(10t)$

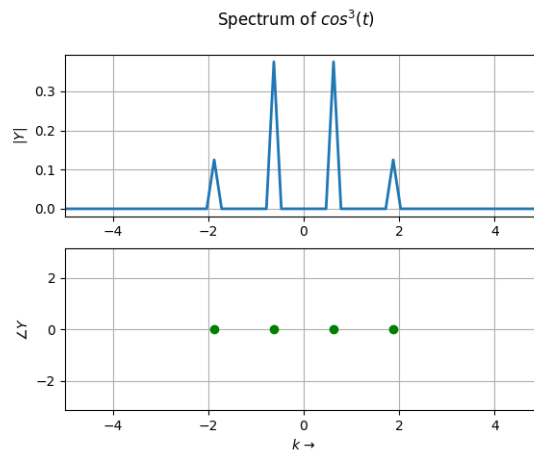
Figure (??) is in line with the theory we have discussed above.

5 Spectra of $\sin^3(t)$ and $\cos^3(t)$

DFT Spectrum of $\sin^3(t)$:

Figure 4: Spectrum of $\sin^3(t)$

DFT Spectrum of $\cos^3(t)$:

Figure 5: Spectrum of $\cos^3(t)$

The above 2 figures are expected because:

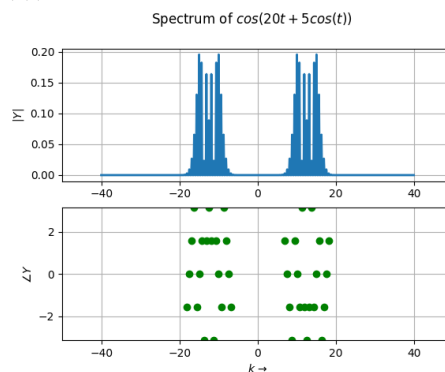
$$\sin^3(t) = \frac{3}{4}\sin(t) - \frac{1}{4}\sin(3t) \quad (3)$$

$$\cos^3(t) = \frac{3}{4}\cos(t) + \frac{1}{4}\cos(3t) \quad (4)$$

So, we expect peaks $\omega = \pm 1 \text{ rad/sec}$ and $\omega = \pm 3 \text{ rad/sec}$.

6 Frequency Modulation with $\cos(20t + 5\cos(t))$

The DFT of $\cos(20t + 5\cos(t))$ can be seen below:

Figure 6: DFT of $\cos(20t + 5\cos(t))$

When we compare this result with that of the Amplitude Modulation as seen in Fig (??), we see

that there are more side bands, and some of them have even higher energy than $\omega = \pm 20 \text{ rad/sec}$.¹

7 DFT of a Gaussian

The DFT of a gaussian is also a gaussian, as shown below:

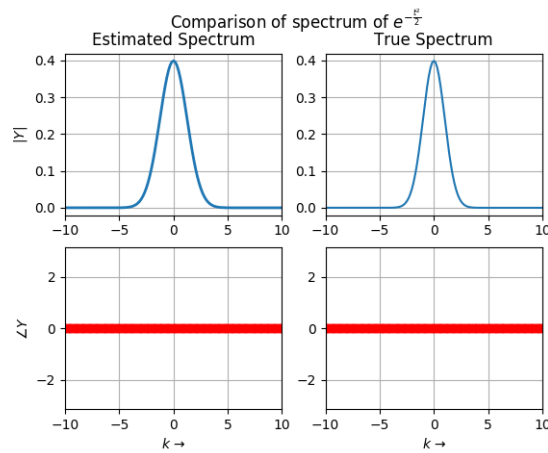


Figure 7: Gaussian Spectrum

8 Conclusion

We have thus found out the DFT's of various signals using the `numpy.fft` package. We need to use the `numpy.fft.fftshift()` and `numpy.fft.ifftshift()` methods to fix distortions in the phase response.

¹J. M. Chowning, *The synthesis of complex audio spectra by means of frequency modulation*, Journal of the Audio Engineering Society, Vol. 21, No. 7, pp. 526-534, 1973